

Applied Physics Assignment 2

Q1) $v(t) = -\omega x_0 \sin(\omega t + \phi)$

$$v_m = \omega x_0$$

$$v(t) = -v_m \sin(\omega t + \phi)$$

at $t=0$ $v = 0.04 \text{ m/s}$, and v_{max} is 0.05 m/s (graph)

$$0.04 = -0.05 \sin(0 + \phi)$$

$$\phi = \sin^{-1} \left[\frac{-0.04}{0.05} \right]$$

$$\phi = -0.9273 \text{ rad}$$

Q2) at mean point mechanical energy = kinetic energy

$$E_k = \frac{1}{2} m v^2$$

$$= \frac{1}{2} (2) (0.85)^2$$

$$= 0.72 \text{ J}$$

$$\begin{array}{l} E_k = 0 \\ E_p = \text{max} \end{array}$$

$$E_k = \text{max}$$

$$E_p = 0$$

for b

$$U(x) = b x^2$$

$$0.5 = b (0.1)^2$$

$$5 \times 10^{-3} \text{ J/cm}^2 = b$$

part a and b answer

Thus:

$$U(x) = 0.72 \text{ J}$$

$$b x^2 = 0.72$$

$$x = \sqrt{\frac{0.72}{5 \times 10^{-3}}}$$

$$x = 12 \text{ cm}$$

Thus the mass turns back before reaching $x = 15 \text{ cm}$ at $x = 12 \text{ cm}$ the mass turns back

Q3a) $F = kx$

$$20 = k(0.2)$$

$$k = 100 \text{ N/m}$$

b) $f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$

$$= \frac{1}{2\pi} \sqrt{\frac{100}{2}}$$

$$= 1.12 \text{ Hz}$$

c) $v = \omega x_0$

$$v = 2\pi(1.12) \times 0.2$$

$$v = 1.4 \text{ m/s}$$

The max speed occur at mean position

d) $a = -\omega^2 x$

$$a = -(2\pi(1.12))^2(0.2)$$

$$a = 9.9 \text{ m/s}^2 \quad (\text{ignoring sign as asked for magnitude})$$

The max acceleration occurs at the ends where restoring force is max.

e) Total Energy = $\frac{1}{2} kx_0^2$

$$= \frac{1}{2} \times 100 \times (0.2)^2$$

$$= 2 \text{ J}$$

$$f) \omega = 2\pi f$$

$$= 2\pi (1.12)$$

$$= 7.037 \text{ rad/s}$$

$$x = \frac{0.2}{3}$$

$$= 0.0666$$

$$\approx 0.0667$$

$$V = \omega \sqrt{A^2 - x^2}$$

$$V = 7.037 \sqrt{0.2^2 - 0.0667^2}$$

$$V = 1.32 \text{ m/s}$$

$$g) a = -\omega^2 x$$

$$a = -(7.07)^2 \times 0.0667$$

$$a = -3.33 \text{ m/s}^2$$

$$a \approx 3.33 \text{ m/s}^2$$

$$Q4a) \omega = \sqrt{\frac{k}{m}}$$

$$(2\pi f)^2 = \left(\sqrt{\frac{k}{m}}\right)^2$$

$$(2\pi(3))^2 \times (1450) = k$$

$$515193 \text{ Nm}^{-1} = k$$

each spring

$$k = \frac{515193}{4}$$

$$k = 128798 \text{ Nm}^{-1}$$

$$b) f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{515193}{1450 \times (73 \times 5)}} = 2.69 \text{ Hz}$$

$$Q5) \omega = \sqrt{\frac{k}{m}}$$

$$2\pi f = \sqrt{\frac{2(7580)}{0.245}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{15160}{0.245}}$$

$$f = 39.5 \text{ Hz}$$

$$Q6) L = 0.45 \text{ m} \quad k = 120 \text{ N/m}$$

$$F_w = W \sin \theta$$

$$= 14 \sin 40$$

$$= 8.99 \text{ N}$$

at equilibrium

$$F_{\text{spring}} = F_w$$

$$kx = 8.99$$

$$120x = 8.99$$

$$x = 0.0749 \text{ m}$$

$$\text{Distance from the top} = 0.45 + 0.0749 = 0.5249 \text{ m}$$

$$b) T = 2\pi \div \sqrt{\frac{k}{m}}$$

$$= 2\pi \div \sqrt{\frac{120 \cos 40}{1.43}}$$

$$= 0.783 \text{ s}$$

$$Q7) x = x_m \cos\left(\frac{2\pi t}{T} + \phi\right)$$

$$\text{Particle 1} \Rightarrow x = \frac{A}{2} \cos\left[\frac{2\pi(0.5)}{1.5} + 0\right] = -0.25 A$$

$$\text{Particle 2} \Rightarrow x = \frac{A}{2} \cos\left[\frac{2\pi(0.5)}{1.5} + \frac{\pi}{6}\right] = -0.433 A$$

$$\begin{aligned} \text{Difference} &= p_1 - p_2 \\ &= -0.25 A - (-0.433 A) \\ &= +0.18 A \end{aligned}$$

$$b) v = -\omega x_m \sin(\omega t + \phi)$$

$$\text{Particle 1} \Rightarrow v = \frac{-2\pi}{1.5} \left[\frac{A}{2} \sin\left[\frac{2\pi(0.5)}{1.5} + 0\right] \right] = -1.813 A$$

$$\text{Particle 2} \Rightarrow v = \frac{-2\pi}{1.5} \left[\frac{A}{2} \sin\left[\frac{2\pi(0.5)}{1.5} + \frac{\pi}{6}\right] \right] = -1.04 A$$

Both are -ve so are travelling in same direction.

Q8) Q7 and Q8 are same

Q9a) • Position and velocity can be in the same direction.

b) • Velocity and acceleration signs are different so can't be in same direction

c) • Acceleration and Position can never be in the same direction they are always opposite to each other.

c) If the mass is doubled and amplitude remains unchanged then there will be no effect on total energy as it is not dependent on mass. $E_T = \frac{1}{2} k x_0$

dii) If we talk about $KE = \frac{1}{2} m v^2$ and $GPE = mgh$ then yes they are dependent on mass.

However with spring constant the ~~KE~~ $PE = \frac{1}{2} k x_{\max}^2$ is not dependent on mass but kinetic energy remains dependent on mass.

Q10a) $x = x_0 \cos(\omega t + \phi)$

$$x = 5 \cos\left(2t + \frac{\pi}{6}\right)$$

at $t = 0$

$$x = 5 \cos\left(\frac{\pi}{6}\right)$$

$$x = 4.33 \text{ cm}$$

b) $v = -\omega x_0 \sin(\omega t + \phi)$

$$v = -2(5) \sin\left(0 + \frac{\pi}{6}\right)$$

$$v = -5 \text{ cm/s}$$

c) $a = -\omega^2 x_0 \cos(\omega t + \phi)$

$$a = -2^2(5) \cos\left(0 + \frac{\pi}{6}\right)$$

$$a = -17.32 \text{ cm/s}^2$$

d) Amplitude = 5 cm

Period $\Rightarrow T = \frac{2\pi}{\omega}$

$$= \frac{2\pi}{2}$$

$$= \pi \text{ Sec}$$

Q11) $m = 2.2 \text{ kg}$ $k = 250 \text{ N/m}$ $T = 0.615 \text{ s}$

a) $\omega = \sqrt{\frac{k}{m}}$

$$\frac{2\pi}{T_n} = \sqrt{\frac{k}{m}}$$

$$T_n = 2\pi \sqrt{\frac{m}{k}}$$

$$T_n = 2\pi \sqrt{\frac{250}{2.2}}$$

$$T_n = 0.589 \text{ s}$$

as T_n is not equal to T ($0.589 \neq 0.615$) the oscillation is damped.

for b

$$\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

$$\left(\frac{2\pi}{T}\right)^2 = \left[\sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}\right]^2$$

$$b^2 = \left[\frac{k}{m} - \frac{4\pi^2}{T^2}\right] \times 4m^2$$

$$\sqrt{b^2} = \sqrt{\left(\frac{250}{2.2} - \frac{4\pi^2}{0.615^2}\right)} \times 4(2.2)^2$$

$$b = 13.38 \text{ Ns/m}$$

b) Damping Ratio (ζ)

$$\begin{aligned}\zeta &= \frac{b}{2\sqrt{mk}} \\ &= \frac{13.38}{2\sqrt{2.2(250)}} \\ &= 0.285\end{aligned}$$

The oscillation is undamped as $0 < \zeta < 1$

Q12) $F_x = -kx$

$$k = 580 \text{ N/m}$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$\omega = \sqrt{\frac{580}{8.35 \times 10^{-28}}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{580}{8.35 \times 10^{-28}}}$$

$$f = 1.326 \times 10^{14} \text{ Hz} \quad \text{Ans}$$

$$m = \frac{1.62 \times 10^{-27}}{2}$$

$$m = 8.35 \times 10^{-28}$$

Q13) $m = 1.5 \text{ kg}$ (tray) $m = 0.275 \text{ kg}$ (ball)

$$k = 185 \text{ N/m}$$

a) Total mass of system = $1.5 + 0.275$
 $= 1.775 \text{ kg}$

$$\omega = \sqrt{\frac{k}{m}}$$

$$\omega = \sqrt{\frac{185}{1.775}} = 10.21 \text{ rad s}^{-1}$$

$$a = -\omega^2 y$$

$$g = -\omega^2 y$$

$$y = \frac{g}{\omega^2}$$

$$y = \frac{9.81}{(10.21)^2}$$

$$y = 0.094 \text{ m}$$

Therefore :- $0.15 + 0.094 = 0.244 \text{ m}$

so the tray will be 0.244 m above point A when the ball leaves the tray.

b) $y(t) = y_0 \cos(\omega t + \phi)$

$$0.094 = 0.15 \cos(10.21t + 0)$$

$$10.21t = \cos^{-1} \left[\frac{0.094}{0.15} \right]$$

$$t = 0.0875 \text{ sec} \quad \text{Ans}$$

c) Total System Energy = PE at max

$$= \frac{1}{2} k y_0^2$$

$$= \frac{1}{2} (185) (0.15)^2$$

$$= 2.08 \text{ J}$$

When Ball leave PE:

$$PE_{\text{leave}} = \frac{1}{2} k y_a^2$$

$$= \frac{1}{2} (185) (0.094)^2 = 0.817 \text{ J}$$

$$\begin{aligned}
 KE \text{ at point of leave} &= \text{Total Energy} - PE_{\text{leave}} \\
 &= 2.08 - 0.817 \\
 &= 1.263 \text{ J}
 \end{aligned}$$

$$KE = \frac{1}{2}mv^2$$

$$1.263 = \frac{1}{2}(0.275)v^2$$

$$v = \sqrt{\frac{2(1.263)}{0.275}}$$

$$v = 3.03 \text{ m/s} \quad \text{Ans}$$

$$\textcircled{Q14} \quad x(t) = x_0 \cos(\omega t + \phi)$$

$$x(0) = x_0 \cos(\phi)$$

$$-0.086 = x_0 \cos(\phi) \rightarrow \text{eq 1}$$

$$v(t) = -\omega x_0 \sin(\omega t + \phi)$$

$$v(0) = -\omega x_0 \sin(\phi)$$

$$0.93 = \omega x_0 \sin(\phi) \rightarrow \text{eq 2}$$

$$a(t) = -\omega^2 x_0 \cos(\omega t + \phi)$$

$$a(0) = -\omega^2 x_0 \cos(\phi)$$

$$48 = -\omega^2 x_0 \cos(\phi) \rightarrow \text{eq 3}$$

for Angular frequency

eq 3 and eq 1

$$48 = -\omega^2 x_0 \cos(\phi)$$

$$48 = -\omega^2 (-0.086)$$

$$\omega = \sqrt{\frac{48}{0.086}}$$

$$\omega = 23.62 \text{ rads}^{-1}$$

for frequency

$$\omega = 2\pi f$$

$$f = \frac{\omega}{2\pi}$$

$$f = \frac{23.62}{2\pi}$$

$$f = 3.76 \text{ Hz}$$

b) using eq 1 and eq 2 (x_0 common)

$$x(0) = v(0)$$

$$\frac{-0.086}{\cos(\phi)} = \frac{0.93}{\omega \sin(\phi)}$$

$$\frac{\omega \sin(\phi)}{\cos(\phi)} = \frac{0.93}{-0.086}$$

$$\tan(\phi) = \frac{0.93}{-\omega 0.086}$$

$$\phi = \tan^{-1} \left(\frac{0.93}{-(23.62)(0.086)} \right)$$

$$\phi \approx \alpha = 0.429 \text{ rad}$$

Quadrant 2

$$\phi = \pi - 0.429$$

$$\phi = 2.71 \text{ rad}$$

$$\text{or } 155^\circ$$

(accept)

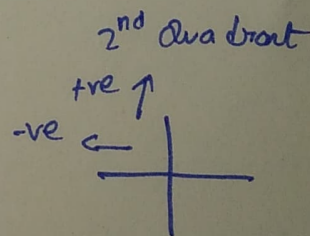
Quadrant 4

$$\phi = 2\pi - 0.429$$

$$\phi = 5.85 \text{ rad}$$

$$\text{or } 335^\circ$$

reject



$$c) -0.086 = \lambda_0 \cos(0.429)$$

$$\lambda_0 = \frac{-0.086}{\cos(2.71)} = 7.263$$

$$\lambda_0 = 0.094 \text{ m}$$

$$1.263 = \frac{1}{2} (8.428) = 4$$

$$V = \sqrt{\frac{2(1.263)}{0.273}}$$

$$V = 3.03 \text{ m/s}$$

(normal of) $\sin$ and $\cos$ given

$$X(0) = V(0)$$

$$X(t) = \lambda_0 \cos(\omega t + \phi)$$

$$X(0) = \lambda_0 \cos(\phi)$$

$$-0.086 = \lambda_0 \cos(\phi)$$

$$\frac{EP \cdot 0}{(\phi) \sin \omega} = \frac{280 \cdot 0 - (\phi) 200}{\cos(\phi)}$$

$$\frac{EP \cdot 0}{280 \cdot 0 - (\phi) 200} = \frac{(\phi) \pi 2 \omega}{\cos(\phi)}$$

$$\frac{EP \cdot 0}{280 \cdot 0 - (\phi) 200} = \tan(\phi)$$

$$\frac{EP \cdot 0}{280 \cdot 0 - (\phi) 200} = \tan^{-1} \left(\frac{EP \cdot 0}{280 \cdot 0 - (\phi) 200} \right)$$

$$\tan^{-1} \left(\frac{EP \cdot 0}{280 \cdot 0 - (\phi) 200} \right) = \phi$$

$$a(t) = -\omega^2 \lambda_0 \cos(\omega t + \phi)$$

$$a(0) = -\omega^2 \lambda_0 \cos(\phi)$$

$$48 = -\omega^2 \lambda_0 \cos(\phi)$$

$$338$$

object

$$EP \cdot 0 - \pi = \phi$$

$$EP \cdot 0 - \pi = \phi$$

$$EP \cdot 0 - \pi = \phi$$

$$122$$

(accept)